Ultrasound, also known as sonography, is a medical imaging technique that uses high-frequency sound waves to create images of structures inside the body. Unlike X-rays or CT scans, ultrasound does not use ionizing radiation, which makes it a very safe and widely used diagnostic tool. The sound waves are emitted by a handheld device called a transducer, which is placed on the skin. These waves travel into the body, bounce off internal organs and tissues, and return to the transducer. A computer then processes these echoes to form real-time images that can be viewed on a monitor.

Ultrasound is commonly used to examine organs such as the liver, kidneys, gallbladder, pancreas, heart, thyroid, and reproductive organs. It is also essential in monitoring pregnancy, as it allows visualization of the developing fetus. Additionally, ultrasound can be used to assess blood flow (Doppler ultrasound), guide medical procedures such as biopsies, and evaluate soft tissue structures like muscles and tendons.

Preparation for an ultrasound examination depends on the specific type of scan being performed, but there are general guidelines that most patients should follow.

Clothing is an important consideration. Patients are usually advised to wear comfortable, loose-fitting clothing that can be easily adjusted or removed depending on the area being examined. For example, if the abdomen is being scanned, the patient may need to lift their shirt or change into a medical gown. Tight clothing can make it more difficult to access the area and may cause unnecessary discomfort. In many cases, patients are asked to remove clothing from the region being examined and are provided with a gown to ensure both accessibility and privacy.

Regarding jewelry and metal objects, it is generally recommended that patients remove any items that might interfere with the examination. This includes bracelets, watches, necklaces, piercings, or any metallic accessories located near the area of interest. While ultrasound itself is not affected by metal in the same way as MRI, these objects can physically obstruct access to the skin or interfere with proper placement of the transducer. For example, bracelets should be removed if the wrist or arm is being examined, and necklaces should be taken off for neck or thyroid scans. It is also advisable to avoid applying lotions, oils, or creams on the skin before the examination, as these can interfere with the contact between the transducer and the skin.

Eating and drinking instructions vary depending on the type of ultrasound. For abdominal ultrasounds, patients are often asked to fast for several hours (typically 6–8 hours) before the examination. This is because food and drink can cause gas to build up in the intestines, which can block sound waves and make it more difficult to obtain clear images. Fasting also ensures that organs like the gallbladder are properly distended and easier to evaluate.

In contrast, for pelvic ultrasounds (especially in women), patients are often instructed to drink several glasses of water before the procedure and avoid urinating until after the scan. A full bladder acts as a “window” that improves visualization of pelvic organs such as the uterus and ovaries. The amount of water and timing will usually be specified by the healthcare provider.

For other types of ultrasound, such as thyroid, musculoskeletal, or vascular studies, no special dietary preparation is usually required. Patients can typically eat and drink normally unless instructed otherwise.

Medications are generally not restricted, and patients should continue taking their prescribed medications unless specifically advised otherwise by their doctor. However, it is always a good idea to inform the healthcare provider about any medications being taken, as well as any relevant medical conditions.

On the day of the examination, patients should arrive on time and follow any specific instructions given by their healthcare provider. During the procedure, a clear gel is applied to the skin to help transmit sound waves efficiently. The transducer is then moved over the area to capture images. The procedure is usually painless, although slight pressure may be applied, which can cause mild discomfort in sensitive areas.

After the ultrasound, patients can typically resume normal activities immediately. There are no side effects from the procedure itself, and results are usually reviewed by a radiologist and sent to the referring physician.

In summary, ultrasound is a safe, non-invasive, and highly useful diagnostic tool. Proper preparation—such as wearing appropriate clothing, removing metal objects, and following dietary instructions—helps ensure that the examination is as effective and accurate as possible.